

Unit-V: AI IN ROBOTICS

❖ What is Robotic Perception?

Robotic perception is the ability of a robot to **sense, interpret, and understand its environment** using data from sensors. It acts like human senses (vision, hearing, touch), enabling robots to make intelligent decisions.

Key Components of Robotic Perception

1. Sensors (Data Collection)

Robots use various sensors to capture environmental data:

- **Vision Sensors** – Cameras (RGB, depth cameras)
- **Distance Sensors** – LiDAR, ultrasonic sensors
- **Inertial Sensors** – IMU (accelerometer + gyroscope)
- **Touch Sensors** – Force and tactile sensors

👉 Example: A self-driving car uses LiDAR + cameras to “see” roads and obstacles.

2. Data Processing

Raw sensor data is noisy and unstructured. It must be processed using algorithms:

- Filtering (noise removal)
- Feature extraction (edges, shapes, objects)
- Signal processing

3. Perception Algorithms

These algorithms help interpret the environment:

- **Computer Vision** (object detection, image recognition)
- **Machine Learning / Deep Learning**
- **Pattern Recognition**

👉 Example: Identifying a pedestrian from camera input.

4. Sensor Fusion

Combining data from multiple sensors to improve accuracy.

- Camera + LiDAR + IMU → better understanding
- Reduces uncertainty and errors

5. Environment Representation

The robot builds a model of its surroundings:

- Maps (2D/3D)
- Object locations
- Obstacles



Types of Robotic Perception

1. Visual Perception

- Uses cameras
- Detects objects, faces, colors, motion

2. Spatial Perception

- Understands distance, depth, and geometry
- Uses LiDAR, stereo vision

3. Auditory Perception

- Uses microphones
- Recognizes sound or voice commands

4. Tactile Perception

- Uses touch sensors
- Detects pressure, texture, and force



Applications of Robotic Perception

- 🚗 **Autonomous Vehicles** – Detect roads, traffic, pedestrians
- 🏭 **Industrial Robots** – Object recognition and assembly
- 🤖 **Service Robots** – Navigation and human interaction
- 🚁 **Drones** – Obstacle avoidance and mapping
- 🏥 **Medical Robots** – Image-guided surgery



Challenges in Robotic Perception

- Noise and uncertainty in sensor data
- Changing environments (lighting, weather)
- Real-time processing requirements
- Occlusion (objects blocking view)
- High computational cost

❖ What is Localization?

Localization is the process by which a robot determines **its position and orientation (pose)** within an environment.

👉 In simple terms:

“Where am I?” is the core question localization answers.

A robot’s pose is usually represented as:

- **Position** → (x, y, z)
- **Orientation** → angle (θ) or rotation

🔍 Why is Localization Important?

Localization is essential because a robot cannot:

- Navigate
- Avoid obstacles
- Plan paths

without knowing its current position.

👉 Example: A delivery robot must know its location inside a building to reach the correct room.

⚙️ Types of Localization

1. Relative Localization (Dead Reckoning)

- Uses internal sensors like:
 - Wheel encoders
 - IMU (accelerometer, gyroscope)
- Estimates position based on previous movement

✅ Simple and fast

❌ Errors accumulate over time (drift)

2. Absolute Localization

- Uses external references:
 - GPS (outdoor)
 - Landmarks (indoor)
- Provides global position

✅ More accurate

❌ May fail indoors or in obstructed areas

3. Probabilistic Localization

- Uses probability to handle uncertainty
- Common methods:
 - **Kalman Filter**
 - **Particle Filter (Monte Carlo Localization)**

✓ Robust and widely used

✗ Computationally intensive

🧠 Key Techniques in Localization

1. GPS-Based Localization

- Used in outdoor robots (e.g., autonomous cars)
- Accuracy depends on signal quality

2. Landmark-Based Localization

- Uses known features (walls, markers, objects)
- Matches sensor data with map

3. Vision-Based Localization

- Uses cameras and computer vision
- Example: Visual odometry

4. LiDAR-Based Localization

- Uses laser scanning to detect surroundings
- Creates precise maps

5. SLAM (Simultaneous Localization and Mapping)

- Robot builds a map **and** localizes itself at the same time

👉 Very important in unknown environments

🔄 Localization Process (Steps)

1. **Sense** → Collect data from sensors
2. **Predict** → Estimate new position based on motion
3. **Update** → Correct estimate using observations
4. **Refine** → Reduce uncertainty

⚠️ Challenges in Localization

- Sensor noise and errors
- Dynamic environments (moving objects)
- GPS signal loss indoors
- Ambiguous or repetitive environments
- Computational complexity

🕒 Applications

- 🚗 Autonomous vehicles
- 🏠 Warehouse robots
- 🚁 Drones navigation
- 🏥 Hospital delivery robots
- 🏠 Home cleaning robots

❖ What is Mapping (Configuring Space)?

Mapping in robotics is the process of **creating a representation (map) of the environment** so that a robot can understand and interact with space.

👉 “Configuring space” means:

- Dividing the environment into meaningful structures
- Identifying **free space, obstacles, and boundaries**
- Organizing it in a way that supports navigation and planning

🧠 Why Mapping is Important

A robot needs a map to:

- 🕒 Navigate from one point to another
 - 🚧 Avoid obstacles
 - 📍 Work with localization (“Where am I on the map?”)
 - 🏠 Perform tasks like cleaning, delivery, inspection
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Types of Maps in Robotics

1. Metric Maps

- Represent environment using precise measurements
- Example: coordinates, distances

Types:

- **Occupancy Grid Map**
 - Space divided into grid cells
 - Each cell: occupied / free / unknown

2. Topological Maps

- Represent environment as a graph
- Nodes = locations, Edges = paths
- ✓ Memory efficient
- ✗ Less precise than metric maps

3. Semantic Maps

- Add meaning to objects
- Example:
 - “This is a door”
 - “This is a table”

👉 Used in intelligent robots and AI systems

Techniques for Mapping

1. Sensor-Based Mapping

Robots use sensors such as:

- LiDAR → distance measurement
- Cameras → visual mapping
- Ultrasonic sensors → obstacle detection

2. SLAM (Simultaneous Localization and Mapping)

- Builds a map while tracking robot position
- Combines:
 - Mapping + Localization

👉 Widely used in autonomous robots

3. 3D Mapping

- Creates detailed 3D models of environment
- Uses:
 - Depth cameras
 - LiDAR point clouds

Mapping Process (Configuring Space)

1. **Data Collection**
 - Gather sensor data
2. **Feature Extraction**
 - Identify walls, objects, edges
3. **Space Representation**
 - Convert into grids, graphs, or models
4. **Map Updating**
 - Continuously refine map

Representation of Space

Occupancy Grid Example:

- Each cell represents:
 - 1 → Occupied
 - 0 → Free
 - ? → Unknown

 Helps robots decide safe paths

Challenges in Mapping

- Noisy sensor data
- Dynamic environments (moving objects)
- Large-scale environments
- Memory and computation limits
- Loop closure problem (recognizing previously visited places)

Applications

-  Autonomous robots (cleaning, delivery)
-  Self-driving cars
-  Industrial automation
-  Drone navigation
-  Service robots in hospitals

❖ What Does “Planning Uncertain Movements” Mean?

In real-world robotics, a robot never has perfect information.
So, **planning uncertain movements** means:

👉 Designing paths and actions **when there is uncertainty in sensing, environment, or motion.**

Sources of uncertainty:

- Sensor noise (wrong readings)
- Actuator errors (wheel slip, drift)
- Unknown or changing environment
- Incomplete maps

Why It Is Important

Without handling uncertainty, a robot may:

- Crash into obstacles
- Lose its path
- Fail to reach the goal

👉 Therefore, robots must **plan robustly**, considering possible errors.

Key Concepts

1. Deterministic vs Probabilistic Planning

Type	Description
Deterministic	Assumes perfect knowledge
Probabilistic	Accounts for uncertainty

👉 Real robots use **probabilistic planning**

2. Belief State

Instead of a single position, the robot maintains:

- A **probability distribution** over possible states

👉 Example: "I am likely here (80%), but maybe elsewhere (20%)."

3. Motion Uncertainty

Even if a robot commands:

- "Move forward 1 meter"

It may:

- Move 0.9 m or 1.1 m

🔍 Techniques for Planning Under Uncertainty

1. Markov Decision Process (MDP)

- Used when state is fully observable
- Components:
 - States
 - Actions
 - Transition probabilities
 - Rewards

👉 Helps choose optimal action under uncertainty

2. Partially Observable Markov Decision Process (POMDP)

- Used when robot **cannot fully observe environment**
- Works with belief states
- ✓ Very powerful
- ✗ Computationally complex

3. Probabilistic Roadmaps (PRM)

- Builds a graph of possible paths
- Considers uncertainty in obstacles

4. Rapidly-exploring Random Trees (RRT)

- Randomly explores space to find paths
- Works well in complex environments

5. Kalman Filter & Particle Filter

- Used to estimate robot state under uncertainty
- Improve planning accuracy

Planning Process Under Uncertainty

1. **Sense** → Gather uncertain data
2. **Estimate** → Predict possible states (belief)
3. **Plan** → Choose safest/optimal path
4. **Act** → Execute movement
5. **Update** → Correct based on feedback

Challenges

- High computational cost
- Real-time decision making
- Dynamic environments
- Trade-off between safety and efficiency

Applications

-  Autonomous vehicles (handling traffic uncertainty)
-  Warehouse robots (dynamic obstacles)
-  Drones (wind disturbances)
-  Medical robots (precision under uncertainty)

❖ Dynamics and Control of Movement

◆ 1. Introduction

Dynamics and control of movement is a core concept in robotics that deals with:

- How robots **move under forces and torques** (Dynamics)
- How robots **control that movement accurately** (Control)

👉 In simple terms:

- **Dynamics = Physics of motion**
- **Control = Making motion behave as desired**

◆ 2. Types of Motion in Robotics

Robotic motion is broadly classified into:

◆ 1. Translational Motion

- Movement in straight line
- Example: Robot moving forward/backward

◆ 2. Rotational Motion

- Movement around an axis
- Example: Robot arm rotating

◆ 3. Combined Motion

- Both translation + rotation
- Example: Robotic manipulator

◆ 3. Dynamics of Movement

Dynamics studies **forces causing motion** using principles from Classical Mechanics

◆ 3.1 Kinematics vs Dynamics

Feature	Kinematics	Dynamics
Focus	Motion only	Motion + forces
Considers	Position, velocity	Force, torque, mass
Example	Path planning	Motor force calculation

◆ 3.2 Types of Dynamics

◆ A. Forward Dynamics

- Input: Forces/torques
- Output: Motion (acceleration, velocity)

👉 “If I apply this force, how will the robot move?”

◆ B. Inverse Dynamics

- Input: Desired motion
- Output: Required forces/torques

👉 “What force is needed to achieve this motion?”

◆ 3.3 Equations of Motion

Robot motion is governed by Newton’s laws:

- Force = Mass × Acceleration
- Torque = Inertia × Angular Acceleration

These are modeled using:

- Differential equations
- Matrix formulations (for multi-joint robots)

◆ 3.4 Factors Affecting Dynamics

- Mass and inertia
- Gravity
- Friction
- External forces
- Joint constraints

Control ensures the robot moves **accurately, smoothly, and reliably**.

◆ 4.1 Types of Control Systems

◆ A. Open Loop Control

- No feedback
- Simple but less accurate

👉 Example: Timer-based motor control

◆ B. Closed Loop Control (Feedback Control)

- Uses sensors for correction
- More accurate and stable

👉 Example: Speed control using encoder

◆ Feedback Mechanism

Steps:

1. Desired motion is set
2. Sensors measure actual motion
3. Error = Desired – Actual
4. Controller corrects error

◆ Controllers Used in Robotics

◆ PID Controller

- Most commonly used controller
- Uses:
 - Proportional (P)
 - Integral (I)
 - Derivative (D)

👉 Works to minimize error over time

◆ **Advanced Controllers**

- Adaptive Control
- Robust Control
- Model Predictive Control (MPC)
- AI-based control

◆ **Trajectory Control**

- Ensures robot follows a **desired path**
- Important in:
 - Industrial robots
 - Autonomous vehicles

◆ **5. Components Involved**

◆ **Sensors**

- Measure position, speed, force
- Example: Encoders, gyroscopes

◆ **Actuators**

- Produce motion
- Example: Motors, hydraulic systems

◆ **Controller**

- Brain of system
- Computes control signals

◆ **Real-World Example**

👉 **Robotic Arm Picking an Object**

1. Kinematics: Plan path
2. Dynamics: Calculate required torque
3. Control: Adjust movement using feedback
4. Sensors: Ensure accuracy

◆ **Challenges in Dynamics and Control**

- Nonlinear behavior
- Uncertainty in environment
- Time delays
- External disturbances
- High computational complexity

◆ **Applications**

- Industrial automation
- Autonomous robots
- Medical robots (surgery)
- Drones and UAVs
- Space robotics

◆ **Advantages**

- Precise motion control
- Improved efficiency
- Automation of complex tasks
- High repeatability

❖ Ethics and Risks of Artificial Intelligence in Robotics

◆ 1. Introduction

Artificial Intelligence (AI) in robotics enables machines to:

- Perceive environment
- Make decisions
- Act autonomously

However, combining AI with physical robots introduces **serious ethical concerns and risks**, because robots can directly affect the real world.

👉 This topic lies at the intersection of Artificial Intelligence, Robotics, and Ethics

◆ 2. Ethical Issues in AI Robotics

◆ 2.1 Safety and Human Life

- Robots operate in **physical environments**, so failures can cause injury or death
- Example:
 - Industrial robot accidents
 - Self-driving car crashes

◆ 2.2 Accountability and Responsibility

- Difficult to assign blame:
 - Developer
 - Manufacturer
 - User
 - AI system itself

◆ 2.3 Privacy Concerns

- Robots use cameras, sensors, microphones
- Constant data collection may violate privacy

👉 Example:

- Home robots recording personal activities

◆ 2.4 Bias and Discrimination

- AI systems may inherit **bias from training data**
- Leads to unfair decisions

👉 Example:

- Facial recognition errors
- Biased decision-making in hiring robots

◆ 2.5 Loss of Human Control

- Fully autonomous robots may act unpredictably
- Hard to intervene in real-time

👉 Ethical concern:

- Should machines be allowed to make critical decisions?

◆ 2.6 Employment and Economic Impact

- Automation replaces human jobs
- Affects:
 - Manufacturing
 - Logistics
 - Services

👉 Ethical issue: Job displacement and inequality

◆ 2.7 Autonomous Weapons

- AI-powered military robots can make lethal decisions

👉 Ethical concerns:

- Lack of human judgment
- Potential misuse

◆ 2.8 Human-Robot Interaction Ethics

- Emotional attachment to robots
- Robots influencing human decisions

👉 Example:

- Elder care robots replacing human interaction

◆ 3. Risks of AI in Robotics

◆ 3.1 Technical Risks

- System failures
- Software bugs
- Sensor errors

👉 Result:

- Unexpected or dangerous behavior.

◆ 3.2 Security Risks

- Robots can be hacked
- Unauthorized control

👉 Example:

- Hacked drone or autonomous vehicle

◆ 3.3 Unpredictability

- AI systems (especially learning systems) behave unpredictably
- Difficult to fully test all scenarios

◆ 3.4 Over-Reliance on Automation

- Humans depend too much on robots
- Reduced human skills and awareness

◆ 3.5 Ethical Misuse

- Use of robots for:
 - Surveillance
 - Warfare
 - Manipulation

◆ 4. Ethical Principles for AI Robotics

To address these issues, systems should follow:

◆ 1. Beneficence

- Robots should benefit humans

◆ 2. Non-Maleficence

- Avoid harm

◆ 3. Autonomy

- Respect human control

◆ 4. Justice

- Fair and unbiased decisions

◆ 5. Transparency

- Explainable AI decisions

◆ 5. Solutions and Mitigation Strategies

◆ 5.1 Safety Mechanisms

- Emergency stop systems
- Redundant sensors

◆ 5.2 Ethical AI Design

- Bias-free training data
- Transparent algorithms

◆ 5.3 Regulations and Standards

- Government policies
- International AI guidelines

◆ 5.4 Human-in-the-Loop Systems

- Humans supervise critical decisions

◆ 5.5 Cybersecurity Measures

- Encryption
- Secure communication

◆ 6. Real-World Examples

- Self-driving cars (accident liability)
- Medical robots (safety and accuracy)
- Military drones (ethical warfare issues)
- Service robots (privacy concerns)

◆ 7. Advantages of Ethical AI Implementation

- Increased trust
- Safer systems
- Better acceptance of robots
- Sustainable development

◆ 8. Disadvantages / Challenges

- Difficult to enforce globally
- Slows innovation
- Complex decision-making
- Trade-offs between safety and efficiency